

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTECOE504C - PROGRAMMING IN JAVA

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which feature of Java makes it platform independent?
A. Use of pointers B. Compilation to machine code
C. Use of bytecode and JVM D. Use of global variables
- 2 Who invented Java Programming? a) Guido van Rossum b) James Gosling c) Dennis Ritchie d) Bjarne Stroustrup
- 3 The default value of an int instance variable in Java is
A. 1 B. Undefined C. null D. 0
- 4 An object of a class is created using
A. new keyword B. static keyword C. class keyword D. this keyword
- 5 Which method is automatically called when an object is created?
A. constructor B. main() C. finalize() D. static()
- 6 Which of the following members belongs to the class rather than objects? A. Instance variable B. Local variable C. static variable D. Method parameter
- 7 Java supports
A. Multiple inheritance using classes B. Multilevel inheritance using classes
C. Hybrid inheritance using classes D. Multiple inheritance directly using classes
- 8 An interface in Java is used to
A. Provide complete method implementation B. Achieve multiple inheritance
C. Create objects directly D. Store instance variables
- 9 Which component is used to compile, debug and execute the java programs? a) JRE b) JIT c) JDK d) JVM
- 10 The method used to start the execution of a thread is
A. start() B. execute() C. run() D. begin()
- 11 What is the extension of java code files? a) .js b) .txt c) .class d) .java
- 12 Which of the following is not an OOPS concept in Java? a) Polymorphism b) Inheritance c) Compilation d) Encapsulation

Q2. Solve the following.

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|-------------------------------------|---|---|
| A) | Explain in detail JVM,JRE and JDK | 6 |
| B) | List out various various operators in java and explain in deatil | 6 |
| Q3. Solve the following. | | |
| A) | Explain the use of 'static' keyword in java | 6 |
| B) | Write an example Java program to demonstrate Method Overloading and method overriding | 6 |
| Q4. Solve Any Two of the following. | | |
| A) | Explain concept of inheritance in Java with example. | 6 |
| B) | Explain concept of class and object in java | 6 |
| C) | List out the different decision making statements in java and write a program to calculate maximum of three numbers in java | 6 |
| Q5. Solve Any Two of the following. | | |
| A) | Explain exception handling in Java. | 6 |
| B) | Explain types of applets and write a simple Java applet program. | 6 |
| C) | Explain in detail multithreading in java | 6 |
| Q6. Solve Any Two of the following. | | |
| A) | Explain the Abstract Window Toolkit (AWT) in Java. | 6 |
| B) | Explain JDBC architecture. | 6 |
| C) | Write different steps of programming using JDBC | 6 |

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